Tutorial - 2

Q. Construct a CFG for a language over $\{A, B\}$ where

$L1 = \{ a^n b^n \mid n >= 1 \}$

$L2 = \{ a^n b^{2n} \mid n >= 1 \}$

$L3 = \{ ww^T \mid w \in \{a, b\}^* \}$

$L4 = \{ wcw^T \mid w \in \{a, b\}^* \}$

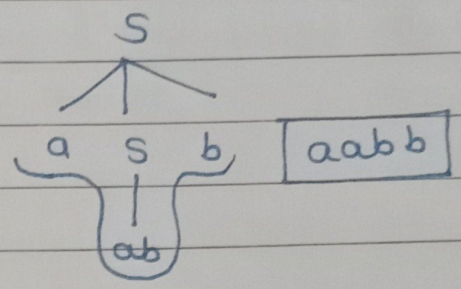
$L5 = \{ a^n b^n c^j \mid n >= 1, j >= 0 \}$

$L6 = \{ c^e a^n b^n \mid n >= 1, e >= 0 \}$

① L1 = ab, aabb, aaabbb ...

$S \rightarrow aSb \mid ab$

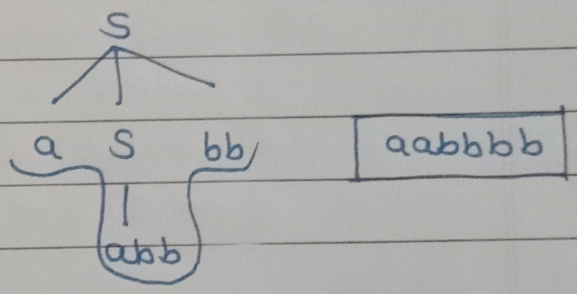
i/p : aabb



② L2 = abb, aabbbb

$S \rightarrow aSbb \mid abb$

i/p : aabbbb



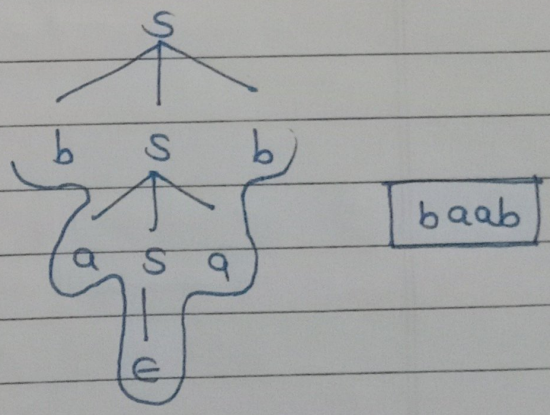
③ L3 = abba, aaaa, baab

$S \rightarrow aSa$

$S \rightarrow bSb$

$S \rightarrow \epsilon$

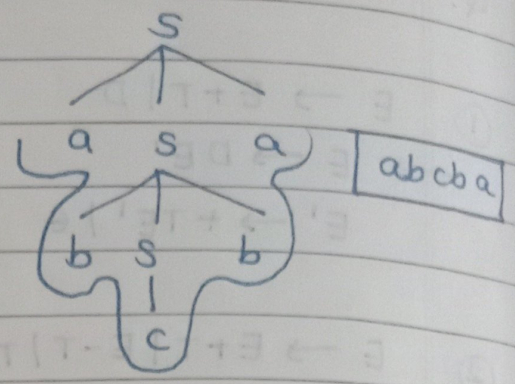
i/p : baab



④ $L_4 = abcba,\ bacab,\ aacaa \ldots$

$S \rightarrow aSa \mid bSb \mid c$

i/p: abcba



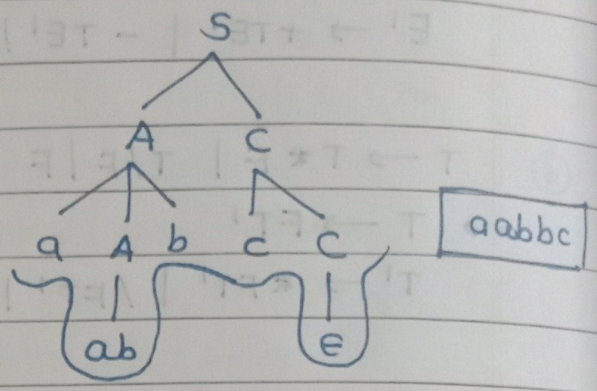
⑤ $L_5 = ab, abc, aabbc \ldots$

$S \rightarrow AC$

$A \rightarrow aAb \mid ab$

$C \rightarrow cC \mid \epsilon$

i/p: aabbc



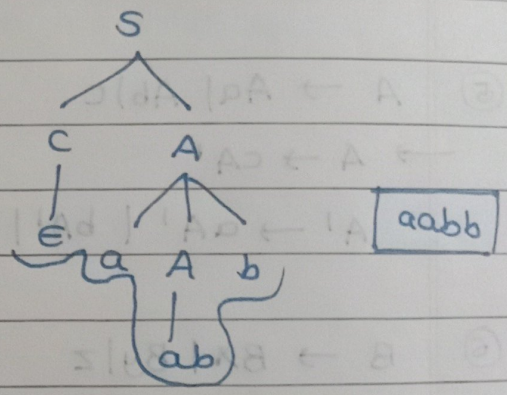
⑥ $L_6 = \{ab,\ cab,\ aabb,\ caabb\}$

$C \rightarrow cA \mid \epsilon$

$A \rightarrow aAb \mid ab$

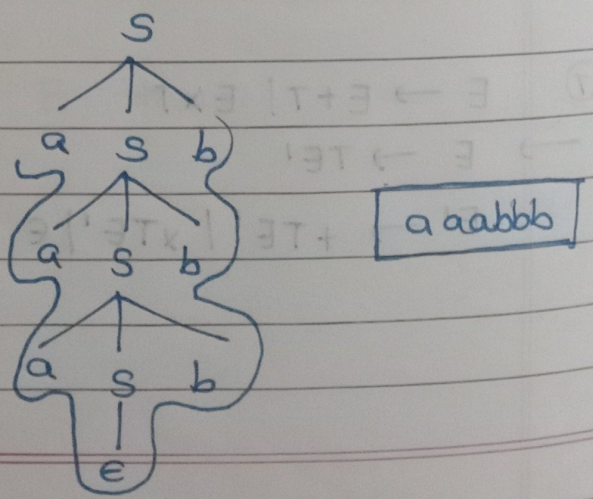
$S \rightarrow CA$

i/p: aabb



⑦ $L_7 = \{a^n \mid b^n \quad n >= 0\}$

$S \rightarrow aSb$

$S \rightarrow \epsilon$

i/p: aaabbb